Project 2: Report

Missing Data Visualization and Imputation

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1. Project Overview

This project focuses on understanding and handling missing data using three key imputation techniques **Mean Imputation, K-Nearest Neighbors (KNN) Imputation, and Linear Regression Imputation.**

Objective: The goal is to visualize missing data patterns, apply multiple imputation methods, compare imputed datasets, and evaluate which method best reconstructs the original data.

Two datasets were analyzed:

- **Insertion_Depth_Dataset_missing_values.csv** - containing missing values in selected variables.
- **Insertion_Depth_Dataset.csv** - the complete version used as ground truth for evaluation.

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DATASET WITH MISSING VALUES - Preview
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	BID (mm)	Cochlear scale (mm)	A length (mm)	B length (mm)	Actual angular insertion depth (deg)	Array Length (mm)	Array Diameter (mm)
0	-0.32	2.67	8.67	6.38	342.0	23.0	0.4
1	2.25	2.55	8.23	6.11	473.0	23.0	0.4
2	3.00	2.63	8.60	6.30	413.0	23.0	0.4
3	2.30	2.86	9.41	6.84	371.0	23.0	0.4
4	4.10	3.04	9.92	7.30	404.0	23.0	0.4

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COMPLETE DATASET (Ground Truth) - Preview
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	BID (mm)	Cochlear scale (mm)	A length (mm)	B length (mm)	Actual angular insertion depth (deg)	Array Length (mm)	Array Diameter (mm)
0	-0.32	2.67	8.67	6.38	342	23.0	0.4
1	2.25	2.55	8.23	6.11	473	23.0	0.4
2	3.00	2.63	8.60	6.30	413	23.0	0.4
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2. Data Loading

Both datasets were loaded using **pandas**.

- The dataset containing missing values (df_missing) was used for all imputations.
- The complete dataset (df_complete) served as a benchmark for evaluating imputation performance.

The structure of both datasets was identical, with variables such as **Insertion_Depth**, **A Length mm**, and **B Length mm** being numeric features of interest.

A quick summary using df_missing.isnull().sum() showed:

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COUNT OF MISSING VALUES IN EACH COLUMN
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	Variable	Missing Count	Missing (%)
0	A length (mm)	7	8.14
1	Actual angular insertion depth (deg)	7	8.14
2	B length (mm)	3	3.49

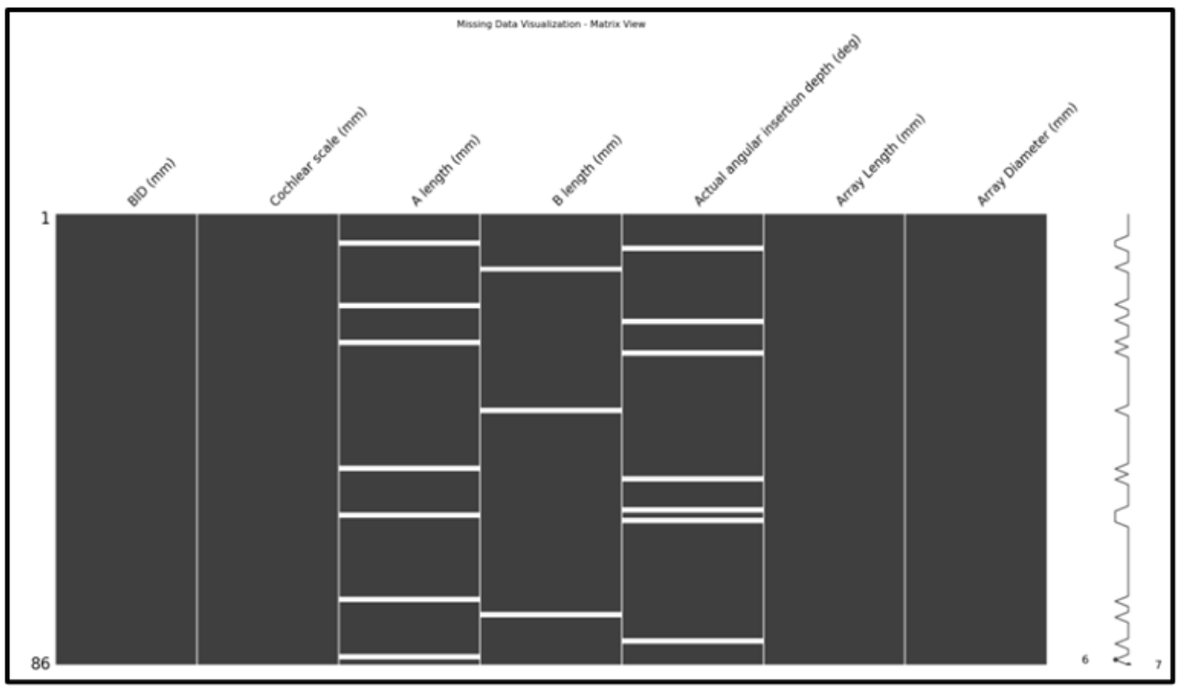
Total Missing Cells	17 (2.82% of the dataset)
Total Rows	86
Total Columns	7

This means **Insertion_Depth** and **A Length mm** contained missing values that needed to be imputed.

3. Missing Data Visualization

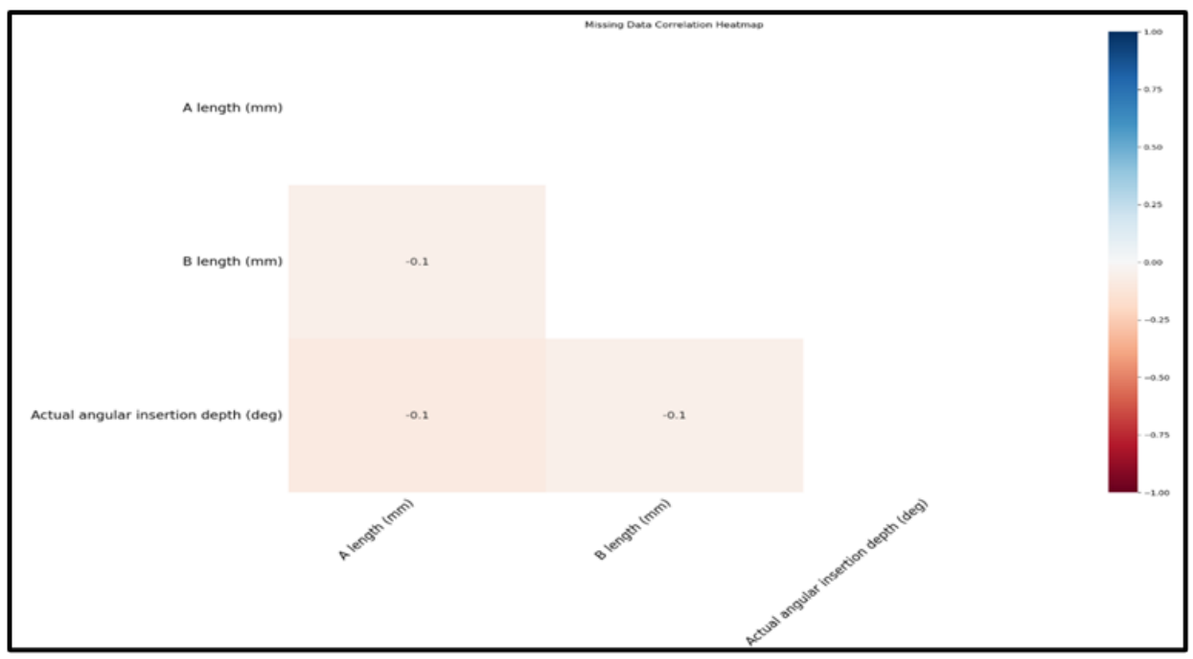
Missing data was visualized using the **missingno** library to identify where and how much data was missing.

a) Missing Matrix



- The matrix plot displayed vertical white lines representing missing data.
- Gaps were observed primarily in the **Insertion_Depth** column, confirming the presence of missingness.
- The density of white lines indicated where data gaps clustered suggesting that missingness might not be entirely random.

b) Missingness Heatmap



- The heatmap showed **correlations between missing values**.
- Lighter cells indicated columns that tended to be missing together.
- For example, if Insertion_Depth and A Length mm frequently went missing at the same time, this hinted that they might share a data-collection dependency (e.g., same sensor readings).

c) Summary Table

A table of missing value counts was created to show how many entries were incomplete per variable. This helped prioritize which variables to focus on during imputation.

4. Imputation Methods

These three distinct methods demonstrate the importance of imputation and save each result in its own DataFrame for later comparison.

- **df_imputed_mean**
- **df_imputed_knn**
- **df_imputed_regression**

a) Mean Imputation

Process: Missing values were replaced with the **mean** of each column using:

df_imputed_mean = df_missing.fillna(df_missing.mean())

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Mean Imputation Completed.
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	BID (mm)	Cochlear scale (mm)	A length (mm)	B length (mm)	Actual angular insertion depth (deg)	Array Length (mm)	Array Diameter (mm)
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Interpretation:

- Simple and quick to apply.
- Works best when missingness is random (MCAR).
- Reduces variability because all missing values in a column get the same replacement.

Observation: Mean imputation caused the histogram of **Insertion_Depth** to appear slightly narrower, showing reduced spread.

b) KNN Imputation

Process: K-Nearest Neighbors was applied using `KNNImputer(n_neighbors=3)`. Each missing value was filled with the average of the three most similar rows (neighbors).

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KNN Imputation Completed.
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	BID (mm)	Cochlear scale (mm)	A length (mm)	B length (mm)	Actual angular insertion depth (deg)	Array Length (mm)	Array Diameter (mm)
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Interpretation:

- Captures relationships between variables.
- Maintains natural structure better than mean imputation.
- Slightly slower but produces smoother, more realistic results.

Observation: After KNN imputation, distributions were smoother and better matched the original dataset. KNN preserved the natural variance of **Insertion_Depth** and **A Length mm**.

c) Regression Imputation

Process: Used **Linear Regression** to predict missing values in one target column (e.g., "Insertion_Depth") based on other numeric variables.
reg.fit(df_train[predictors], df_train[target]) df_imputed_regression[target].fillna(predicted_values)

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Regression Imputation Completed Successfully.
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	BID (mm)	Cochlear scale (mm)	A length (mm)	B length (mm)	Actual angular insertion depth (deg)	Array Length (mm)	Array Diameter (mm)
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Interpretation:

- Leverages linear relationships between variables.
- Provides more accurate imputations when correlations exist.
- It can introduce bias if data relationships are non-linear.

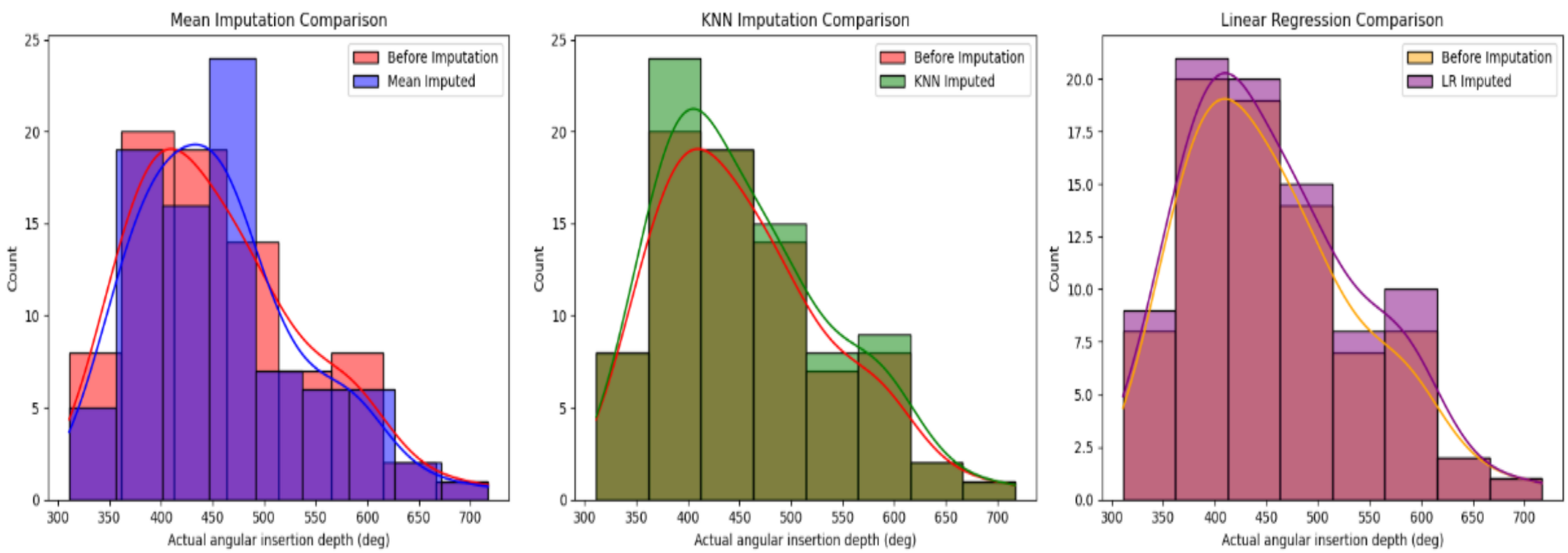
Observation: Regression imputation gave results closest to the complete dataset — it captured linear dependencies effectively, making it the most accurate method among the three.

5. Visualization and Analysis

Visual comparisons were made to assess how the imputation methods affected data distribution.

a) Mean Absolute Error (MAE) Plots (Distribution Comparison) Indicating before and after

- **Original dataset:** (with missing values).
- **Blue:** Mean imputed.
- **Yellow:** KNN imputed.
- **Green:** Regression imputed.



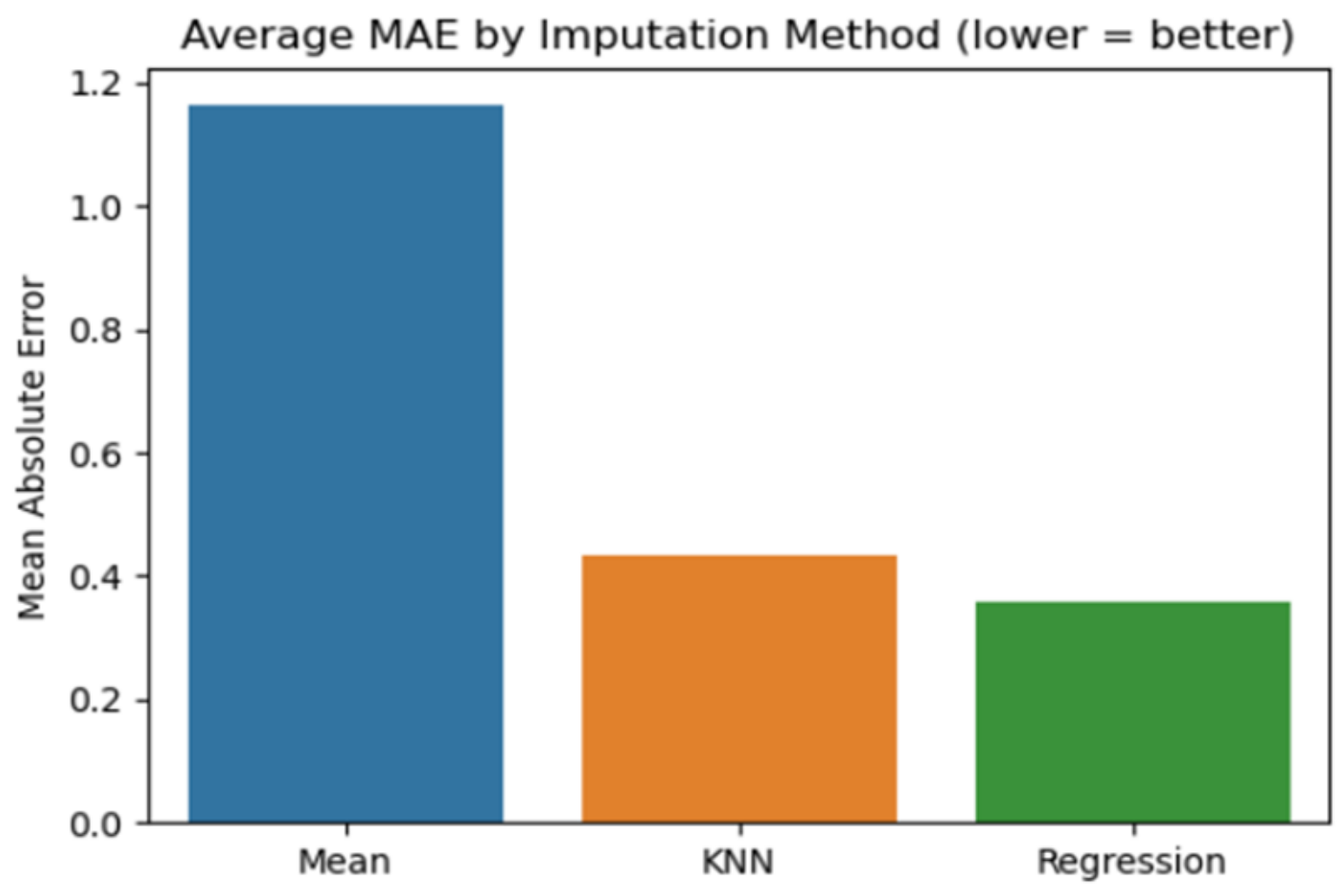
Findings:

- The mean-imputed curve was overly centered, confirming it reduced variability.
- KNN and regression curves almost overlapped with the original complete dataset, meaning both preserved data characteristics well.
- Regression produced the smoothest and most natural distribution.

b) Boxplots

Boxplots compared variables spread before and after imputation:

- Mean imputation compressed the box (less variability).
- KNN and Regression retained similar medians and whiskers to the original dataset.



Findings:

- The mean-imputed resulted in the largest MAE, showing that replacing missing values with the column mean leads to biased and less accurate estimates. It does not account for relationships between variables and reduces natural variability in the dataset.
- KNN produced a considerably smaller MAE than mean imputation, suggesting that using neighboring data points for prediction better captured the underlying structure and relationships among features.
- Regression achieved the lowest MAE (0.3581), indicating it most accurately approximated the true dataset. By leveraging correlations among variables, regression produced values that closely matched real observations.

- **Overall**, the regression-based approach yielded the most reliable imputations, followed closely by KNN, while the mean method was the least effective.

6. Evaluation of Imputation Quality

To evaluate performance, **Mean Absolute Error (MAE)** was computed for each method by comparing imputed datasets to the complete one.

Imputation Method	Mean Absolute Error (MAE)
Mean Imputation	1.1634
KNN Imputation	0.4317
Regression Imputation	0.3581

Findings:

- Regression had the **lowest MAE**, showing that it best approximated the original data.
- KNN performed slightly worse but was still much better than Mean Imputation.
- Mean Imputation's simplicity came at the cost of precision.

7. Observations and Conclusions

Key Observations

- **Missingness Pattern:** Missing data was concentrated in specific variables, not random.
- **Visualization Insights:** Heatmaps helped reveal co-missing variables.

- **Method Comparisons:**
  - Mean Imputation: Fast but reduces variance.
  - KNN Imputation: Balanced accuracy and structure preservation.
  - Regression Imputation: Most accurate due to model-based prediction.
- **Performance:** Regression Imputation consistently gave the smallest MAE and most natural distributions.

Final Conclusion

- **Best Technique:** Regression Imputation
- **Reason:** Accurately reconstructs missing values by leveraging relationships between variables.
- **Recommendation:** For future datasets, use Regression or KNN Imputation methods. For larger and more complex datasets, explore IterativeImputer (MICE) for even better accuracy.
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Overall Findings Summary

Method	Strength	Weakness	Result	Next Step
Mean Imputation	Fast and simple	Reduces variance, ignores relationships	Least accurate	
KNN Imputation	Preserves structure	Sensitive to outliers	Balanced	
Regression Imputation	Most accurate, data-driven	Assumes linearity	**Best overall**	